

HoldMeTolt — Product & Technical Evolution Roadmap (MVP → P1 → V1 → V2)

Document Type: Product & Technical Evolution Roadmap

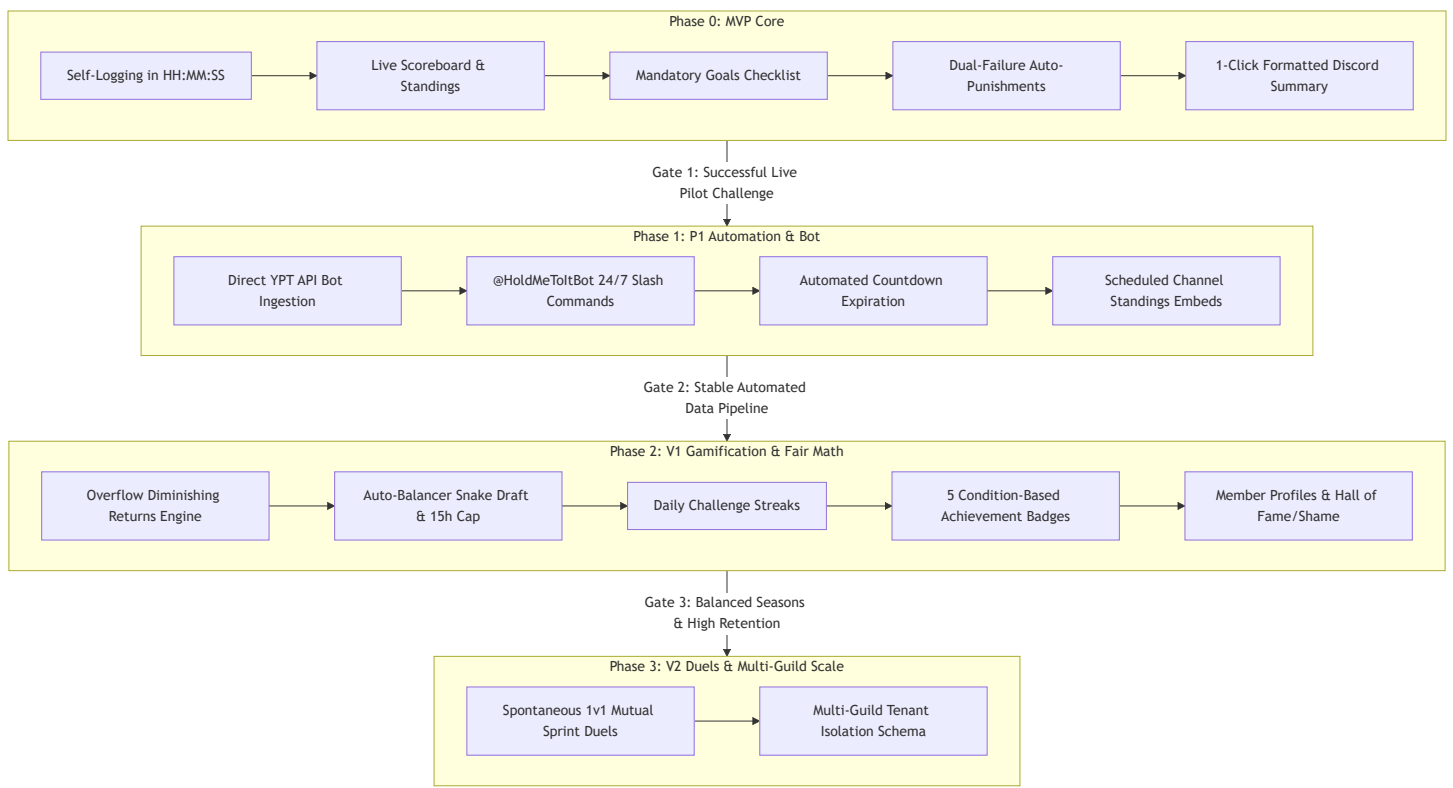
Audience: Engineering Team (3–4 Developers), Community Moderators & Stakeholders

Cadence: Milestone-Gated Evolution (Sprints planned dynamically per phase)

Reference Stack: Next.js (App Router), TypeScript, Tailwind CSS, shadcn/ui, Prisma, PostgreSQL (Supabase/Neon), Auth.js (Discord OAuth), Vitest, Vercel

1. Phased Trajectory & Evolution Pipeline

HoldMeTolt evolves sequentially across four distinct maturity phases. Progress between phases is governed by **Milestone Quality Gates** rather than arbitrary calendar deadlines:



2. Technical Architecture & Component Evolution

The technical architecture is designed to scale modularly as new phases unlock:

HoldMeToIt Architecture Evolution

- └─ Client / Presentation Layer
 - └─ Next.js 14+ (App Router), React, Tailwind CSS, shadcn/ui
 - └─ Phase 0: Participant cockpit, match scoreboard, admin override grid
 - └─ Phase 1: Countdown ticker, live embed previews
 - └─ Phase 2: Profile showcases, badge showcases, draft workbench
 - └─ Phase 3: 1v1 Duel arena modal
- └─ Business Logic & Service Layer
 - └─ Next.js Server Actions & Route Handlers
 - └─ Phase 0: Clock converters (`HH:MM:SS`), deficit pace engine, punishment validator
 - └─ Phase 1: YPT payload validator, countdown state machine
 - └─ Phase 2: Diminishing returns calculator, snake-draft auto-balancer
 - └─ Phase 3: P2P duel challenge handshake service
- └─ Identity & Access
 - └─ Auth.js (Discord OAuth 2.0 Provider – `identify` scope)
 - └─ Session-based RBAC: `SPECTATOR` → `PARTICIPANT` → `ADMIN`
 - └─ Phase 3: Multi-guild permission resolver
- └─ Bot & Daemon Infrastructure (Phase 1+)
 - └─ Node.js Discord.js service (`@HoldMeToItBot`)
 - └─ Inter-service communication via secured bearer token API routes
 - └─ Scheduled task runners for midnight check-ins and final announcements
- └─ Persistence Layer
 - └─ Prisma ORM querying PostgreSQL (Supabase / Neon)
 - └─ Migration-controlled schema updates per phase
- └─ Quality Assurance & Infrastructure
 - └─ Vitest (domain math unit tests & calculation invariants)
 - └─ Vercel Hosting + GitHub Actions CI/CD

3. Phase 0 (MVP Core): The Spreadsheet Exorcism

3.1 Strategic Objective

Eradicate manual host labor. Deliver a robust, reliable web application capable of running a full community study battle from kickoff to punishment resolution with zero reliance on Google Sheets.

3.2 Core Capabilities

1. **Discord OAuth Authentication:** One-click login; automatic avatar and display name sync; fallback spectator mode.
2. **Multi-Format Challenge Creator:** Host-configured battles supporting `TEAM_VS_TEAM` (e.g. *Bees vs Butterflies*), `DUOS` ($N = 2$), and `SOLOS` ($N = 1$).
3. **Pre-Event Declarations:** Mandatory submission of weekly target hours (`HH:MM:SS`) and weekly to-do list tasks; strict lock invariant once the challenge goes `ACTIVE` .
4. **Clock-Time Self-Logging:** Daily hours entry in `HH:MM:SS` format with a 24-hour single-day safety limit.
5. **Head-to-Head Live Scoreboard:** Team totals, current leader, lead margin delta, and unified participant leaderboard.
6. **Dynamic Deficit Tracker:** Real-time catch-up pace indicator showing exact hours/day needed over remaining days.
7. **Dual-Failure Auto-Punishment:** Auto-flags participants who fail their hours target OR leave declared tasks unfinished.
8. **Direct Punishment PFP Download:** Instant one-click download of the punishment asset directly from the web app.
9. **Admin Manual Override Grid:** Host inline-edit capability to adjust any participant's logged hours or unlock tasks in case of emergency.
10. **1-Click Discord Summary Generator:** Formatted markdown text generator ready to copy-paste into `#study-announcements` .

3.3 Database Entities at Phase 0

- `User` (Discord identity, role)
- `Challenge` (title, timestamps, format, status)
- `Team` (name, color, challenge relation)
- `ParticipantEnrollment` (target seconds, status)
- `DailyStudyLog` (date, duration seconds, override flag)
- `WeeklyGoal` (description, completion boolean)

- `PunishmentRecord` (flagged status, pardon notes)

3.4 Phase 0 Milestone Gate (Promotion to P1)

- ☐ Successful deployment to staging/production on Vercel.
- ☐ 100% test coverage on domain calculation formulas (`HH:MM:SS` parsing, deficit pacing, punishment evaluation).
- ☐ **Live Pilot Challenge:** Run 1 full community challenge with real users, using a shadow Google Sheet as a fallback safety net to verify zero calculation divergence.

4. Phase 1 (P1): Automation & Integrations

4.1 Strategic Objective

Eliminate manual self-logging friction by connecting an automated YPT data ingestion bridge, deploying an interactive Discord bot, and automating event lifecycle transitions.

4.2 Core Capabilities

4.2.1 Direct YPT API Bot Ingestion Engine

- **Context:** An independent YPT scraper/bot built by a teammate streams verified study times directly to HoldMeTolt.
- **Ingestion Route (`POST /api/v1/ingest/ypt`):**
 - Authenticated via secure pre-shared Bearer API Token.
 - Payload Schema:

```
{
  "discordId": "3141592653589793",
  "challengeId": "clx...",
  "date": "2026-09-08",
  "durationSeconds": 16200,
  "yptSubject": "Mathematics"
}
```

- Upserts atomic `DailyStudyLog` records and triggers immediate recalculation of team scores and individual pace.

- **Resilience:** If the scraper experiences network hiccups or YPT session expiration, the system allows participants to self-log manually on the web as a fallback.

4.2.2 24/7 @HoldMeToItBot Discord Bot

- **Slash Commands:**
 - `/stats [user]` — Returns rich Discord embed showing user's daily hours, weekly target progress, and goal checklist.
 - `/standings` — Returns current team scores, lead delta, and individual top 3 podium.
 - `/deficit` — Displays remaining hours and required daily pace to stay out of the punishment bracket.
- **Scheduled Automated Broadcasts:**
 - Mid-day check-in embed automatically posted to `#study-announcements`.
 - Final results announcement embed automatically dispatched when the event concludes.

4.2.3 Event Timer & Automated Countdown Expiration

- **Host Manual Kickoff:** Hosts retain explicit control over when an event starts to ensure rosters are finalized.
- **Automated Expiration:** Once the end timestamp expires:
 - System flips challenge status: `ACTIVE` → `COMPLETED`.
 - Freezes all logging forms and inputs immediately.
 - Executes the punishment evaluation routine and updates the Punishment Wall.

4.3 Database Schema Additions (P1)

- `ApiKey` (stores hashed tokens for teammate YPT bot authorization)
- `BotBroadcastLog` (audit records of automated Discord channel embeds sent)

4.4 Phase 1 Milestone Gate (Promotion to V1)

- ☐ Continuous 7-day automated YPT data ingestion without dropouts or duplicate entries.
- ☐ `@HoldMeToItBot` responds to slash commands with latency $< 800\text{ms}$.
- ☐ Challenge auto-completes and accurately flags punishments without host intervention.

5. Phase 2 (V1): Gamification, Retention & Fair Balancing

5.1 Strategic Objective

Introduce advanced competitive fairness mechanics, prevent single-player carry imbalance, reward consistency through daily streaks, and cultivate long-term community pride through member profiles and achievement badges.

5.2 Core Capabilities

5.2.1 Overflow Hours & Diminishing Returns Engine

- **The Problem:** In team battles, one participant studying 16 hours/day when they committed to 4 hours/day creates massive score distortion, demoralizing the competing team.
- **The Solution:** Dual-Credit Scoring Architecture:
 - i. **Personal Profile & Badges:** The grinder receives **100% full credit** for all logged hours. No hard effort is ever discarded or demeaned.
 - ii. **Team Battle Match Score:** Hours beyond the declared daily commitment apply a damping factor ($\alpha = 0.50$) to preserve balanced team competition.
- **Piecewise Mathematical Formula:**

For a day with declared daily target $T_{\text{daily}} = T_{\text{weekly}}/7$ and logged time L :

$$\text{TeamScore}_{\text{day}} = \begin{cases} L & \text{if } L \leq T_{\text{daily}} \\ T_{\text{daily}} + 0.50 \times (L - T_{\text{daily}}) & \text{if } L > T_{\text{daily}} \end{cases}$$

5.2.2 Auto-Balancer Snake Draft & 15-Hour Cap

- **15-Hour Daily Cap:** Declared targets cannot exceed 15 hours/day (105 hours/week) to safeguard member well-being and prevent fraudulent claims.
- **Algorithmic Snake Draft:**
 - Calculates a composite expected capacity for each member:

$$\text{Rating}_i = 0.6 \times \text{DeclaredDailyTarget}_i + 0.4 \times \text{HistoricalDailyAverage}_i$$

- Sorts participants descending by Rating_i and executes an alternating snake draft ($A, B, B, A, B \dots$) to produce rosters with nearly identical total projected hours.
- **Host Override Workbench:** Interactive drag-and-drop roster editor allowing mods to swap participants before confirming the draft.

5.2.3 Daily Challenge Streaks

- **Mechanic:** A participant's daily streak increments by 1 for every challenge day where logged hours $\geq 100\%$ of their daily target pace.
- **Reset Invariant:** If an active challenge day ends with 0 hours logged, the daily streak resets to 0.

5.2.4 Condition-Based Achievement Badges

Badge	Title	Condition Trigger
	Centurion	Accumulate ≥ 100 hours of verified lifetime study time.
	Night Owl	Accumulate ≥ 30 hours between 10:00 PM and 4:00 AM.
	Flawless Grinder	Achieve 100% of target hours AND 100% of weekly goals in an event.
	Comeback Kid	Overcome a > 5 -hour deficit in the final 48 hours to pass target.
	Veteran	Complete 5 community challenge seasons.

5.2.5 Member Profile Cockpit (/profile/[discordId])

- Total lifetime verified study time (HH:MM:SS).
- Team battle Win/Loss record.
- Current active daily streak.
- Unlocked achievement badges showcase.
- **Hall of Accountability Archive:** Public tally of successful completions vs times assigned the Punishment PFP.

5.3 Database Schema Additions (V1)

- UserBadge (relation linking User to BadgeDefinition with earned timestamp)
- StreakRecord (current streak, longest streak, last increment date)
- HistoricalSeasonRecord (archived snapshot of past battle placements and stats)

5.4 Phase 2 Milestone Gate (Promotion to V2)

- ☐ Diminishing returns formula mathematically verified across 50+ simulated battle scenarios.
- ☐ Auto-balancer produces rosters with $< 5\%$ delta in projected team capacity.
- ☐ Profile pages render dynamic user statistics in $< 300\text{ms}$.

6. Phase 3 (V2): Spontaneous Duels & Scale Readiness

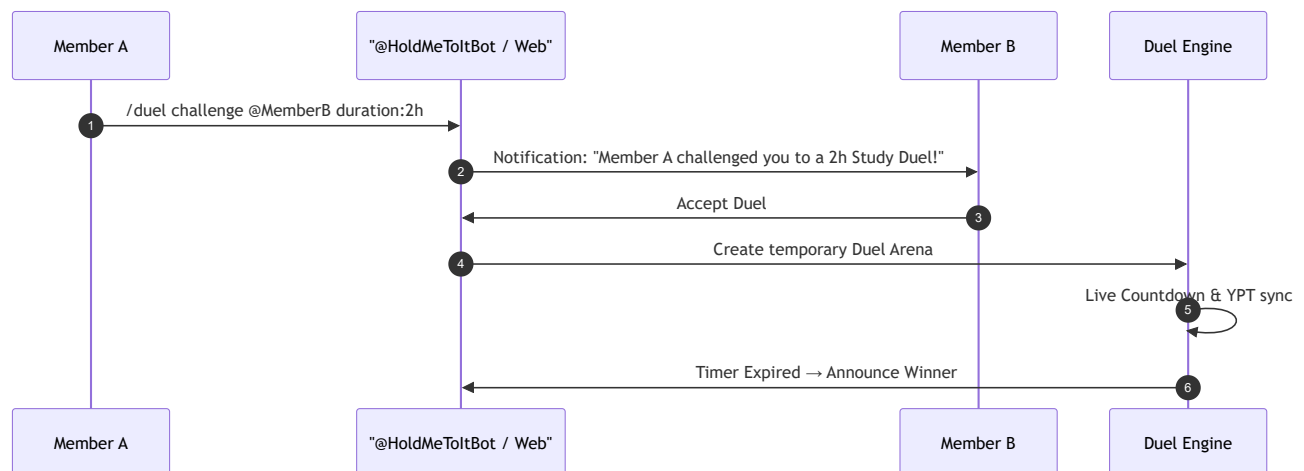
6.1 Strategic Objective

Empower community members to challenge friends to quick study sprints on demand without moderator overhead, while establishing multi-tenant database isolation to allow other Discord communities to onboard.

6.2 Core Capabilities

6.2.1 Spontaneous 1v1 Mutual Study Sprint Duels

- **Concept:** Informal peer-to-peer study sprints (e.g. 1-hour, 2-hour, or single-day duels) initiated between two friends.
- **Lifecycle Flow:**



- **Zero Moderator Burden:** Sprints operate autonomously; no admin approval or management required.

6.2.2 Multi-Guild Database Preparation

- Introduce optional `guild_id` foreign key relations across `Challenge`, `User`, and `Team`.
- Decouples community-specific configurations (e.g. punishment PFP assets, announcement channels), enabling HoldMeTolt to be added to external Discord servers.

6.3 Database Schema Additions (V2)

- `Duel` (`initiator_id`, `opponent_id`, `duration_seconds`, `status`, `winner_id`)
- `GuildConfig` (`guild_id`, `announcement_channel_id`, `admin_role_id`)

7. Master Comparison Across All Phases

Dimension	Phase 0 (MVP)	Phase 1 (P1)	Phase 2 (V1)	Phase 3 (V2)
Study Logging	Manual Web (HH:MM:SS)	Automated YPT Bot + Web	Automated YPT Bot + Web	Automated YPT Bot + Web
Discord Interface	1-Click Clipboard Copy	@HoldMeToItBot Slash Commands	@HoldMeToItBot + Embeds	Bot + /duel Commands
Event Expiration	Host Manual Lock	Automated Timer Expiration	Automated Timer Expiration	Automated Timer Expiration
Team Rostering	Manual Host Assign	Manual Host Assign	Auto-Balancer Snake Draft	Auto-Balancer Snake Draft
Daily Target Cap	Uncapped	Uncapped	15 Hours / Day Cap	15 Hours / Day Cap
Overflow Scoring	100% Linear	100% Linear	Diminishing Returns ($\alpha = 0.5$)	Diminishing Returns
Streaks & Badges	✗	✗	Daily Streaks + 5 Badges	Streaks + Badges
Member Profiles	Basic Dashboard	Basic Dashboard	Full /profile + Hall of Fame	Full /profile + Duel Stats
P2P Study Duels	✗	✗	✗	Mutual 1v1 Sprints
Multi-Server	Single Server	Single Server	Single Server	Multi-Guild Schema

8. Technical Risk Management & Mitigations

Risk	Impact	Likelihood	Technical Mitigation Strategy
Teammate YPT Bot Downtime	High	Medium	Keep manual web self-logging fully functional as an active fallback. The system never blocks user input if the scraper drops.
Discord Rate Limiting	Medium	Low	Queue channel announcements and bot responses; use webhooks with exponential backoff.
Extreme Outlier Skew	High	High	Addressed in Phase 2 via the 15h daily target cap and the 0.50 diminishing returns formula on team scores.
Timer / Input Disputes	Medium	Medium	Mitigated via the Admin Manual Override Grid with <code>is_admin_override</code> audit logging.
Data Loss During Live Battle	Critical	Low	Relational PostgreSQL persistence with atomic database transactions; daily backup snapshots on Supabase/Neon.